

DEVIN WONG

MECHANICAL ENGINEERING AT UNIVERSITY OF CALIFORNIA, SAN DIEGO

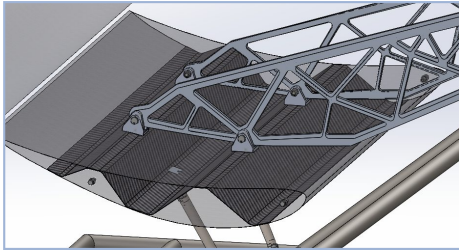
✉ devinwong51@gmail.com

in linkedin.com/in/devinwwong

☎ (408) 290-1237

REAR WING SPAR – FORMULA SAE

TRITON RACING



What?

- Corrugated **800 mm** carbon-fiber rear-wing spar
- Under **10 mm** deflection at **400 N/m**
- Target **FOS 1.5+** under peak load
- Layup: **4-ply 2x2 twill**

How?

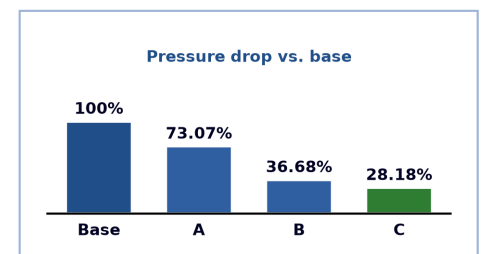
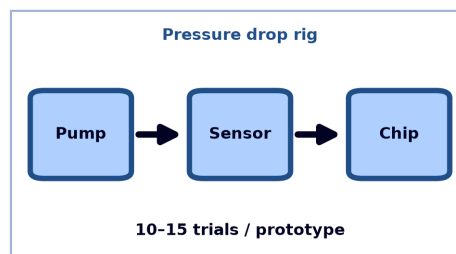
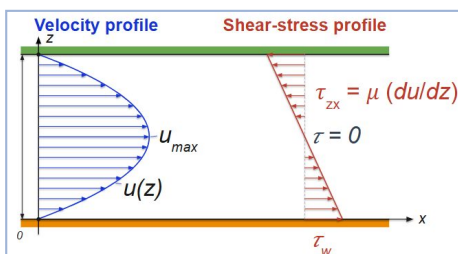
- Layup modeled in **ANSYS ACP**
- Mesh convergence to **2.5% GCI**
- Resolved BC singularities for true stress
- SolidWorks mold, Fusion CAM
- CNC-milled foam, vacuum-bag layup

Results

- Peak deflection **0.38 mm**, on target
- FEA and hand calc in agreement
- Validated **FOS 5.5; 0.5 kg** part
- Mold held to **0.004"** tolerance
- No critical delamination

MICROFLUIDIC CFD & FLOW TESTING – ROCHE

Roche



What?

- Wall-shear uniformity study
- **Four** candidate channel designs
- Metrics: shear and pressure drop

How?

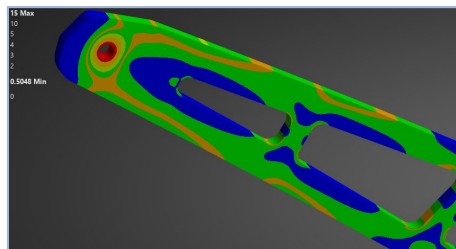
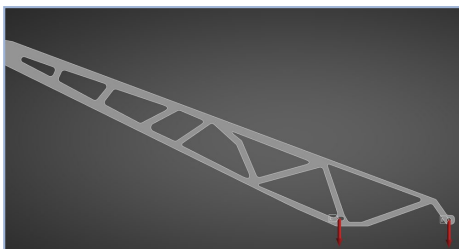
- Steady single-phase **laminar** CFD
- Flow rates set to a common shear
- Grid-independent to **15M** cells
- Pump and pressure-sensor rig

Results

- Cut pressure drop **~72%**
- Cut shear non-uniformity **~70%**
- CFD validated against physical pressure measurements

REAR WING MOUNT ANALYSIS – FORMULA SAE

TRITON RACING



What?

- **6061 aluminum** swan-neck rear-wing mount
- **Topology-optimized** for weight and stiffness
- Target **FOS 1.5+** at **200 N** load

How?

- Hand calc verified against **ANSYS FEA**
- Five-level grid study, **2.2M** elements
- Convergence to **GCI < 0.5%**
- Force-probe and bolt FOS checks
- Fabricated via water jet

Results

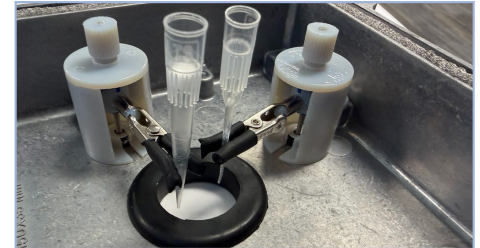
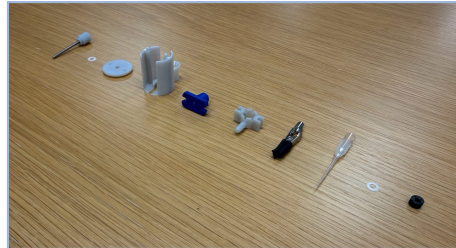
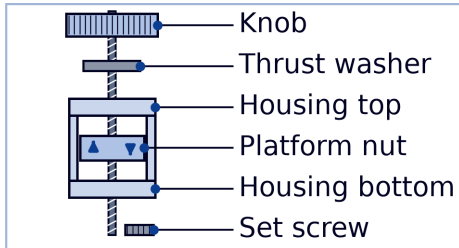
- Weight cut **77%** vs. initial design
- Static bolt margin at critical joint
- Peak deformation **11.43 mm**

DEVIN WONG

MECHANICAL ENGINEERING AT UNIVERSITY OF CALIFORNIA, SAN DIEGO

✉ devinwong51@gmail.com
in linkedin.com/in/devinwwong
☎ (408) 290-1237

PRECISION Z-POSITIONING FIXTURE – ROCHE



What?

- Precise Z-control of probes
- Replacement for manual setup
- Within a **30 mm** height envelope

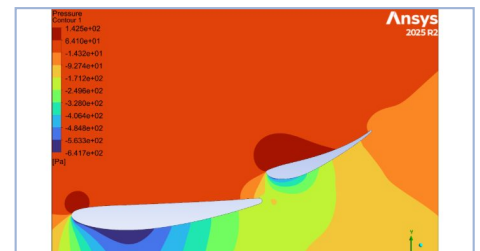
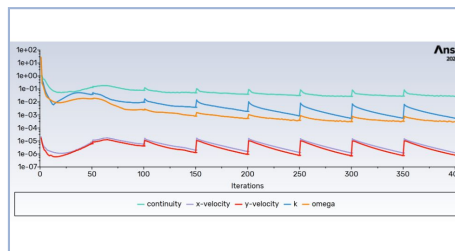
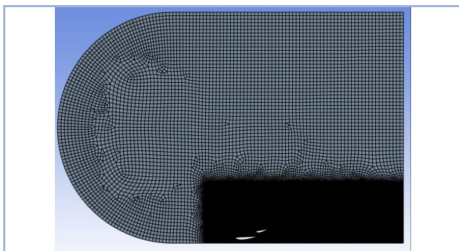
How?

- **Lead-screw** knob-driven travel
- Housing and platform **3D-printed**
- Modeled and iterated in **SolidWorks**

Results

- Fine, repeatable Z positioning
- Eliminated manual position error
- Validated with printed prototype

FRONT WING 2D CFD – FORMULA SAE



What?

- Two-element front-wing CFD study
- Target **200 N** at **15 m/s** speed
- **NACA 4412 + S1223** airfoils
- Minimum **2D Cl** of **2.49**

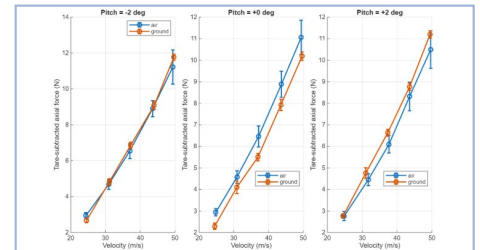
How?

- **2D** CFD in **ANSYS Fluent**
- C-mesh domain, **k- ω SST** model
- Four-level mesh to **872k** cells
- **GCI** via Richardson extrapolation
- **50**-point sweep of **5** parameters

Results

- **Cl 2.74, Cd 0.25, L/D 11.0**
- Met the **2D Cl 2.49** target
- **2.4% GCI** — mesh independent
- Most sensitive to element spacing

WIND TUNNEL VALIDATION – FORMULA SAE



What?

- Physical check on **CFD** trends
- **9%**-scale 3D-printed model
- Tested in **UCSD** wind tunnel
- Trend validation, not 1:1 match

How?

- Air- and ground-supported setups
- **3** pitch angles, **5** flow speeds
- Zero-flow tare per condition
- Matched **CFD** case at **0°** pitch

Results

- Axial force linear in **q**, **$R^2 > 0.996$**
- Matched CFD **V^2** force scaling
- Force ratio agreed to **3.8%**
- Peak downforce nose-down, per CFD